

Predictive Cyber Defence

Technical Evaluator User Guide

SIH internal-round frontend demonstration • deterministic synthetic telemetry • React/TypeScript/Vite/Tailwind

1. What the system does

Predictive Cyber Defence is a SOC-style frontend for a network attack forecasting platform. It combines current risk, future risk windows, predicted attack stages, network-path investigation, feature contributions, MITRE ATT&CK; mapping and alerts. The frontend deliberately separates presentation from inference: the demo consumes centralized mock data through an API-shaped service.

Important limitation: the dashboard does not claim that a live ML model or network sensor is connected.

2. Dashboard overview

Area	Meaning
Current Risk	Latest aggregate risk score; demo value 86% / High Risk.
Forecast Risk	Future risk checkpoints: 63%, 74%, 86%.
Attack Stage	Most likely next stage: Lateral Movement, 82% confidence.
Forecast Horizon	10-second windows across 120 windows.
Active Signals	Suspicious hosts, anomalous flows, high-risk alerts and predicted paths.
Risk Trajectory	Observed history followed by forecast.
Network Graph	Interactive host-to-host topology.
SHAP	Feature contribution explanation.
MITRE	Technique hypotheses and supporting evidence.
Alerts	Investigation queue with filters and status.

3. Meaning of every KPI

Current Risk is the operational summary. Forecast Risk expresses where the risk may move next. Predicted Attack Stage adds semantic context to the score. Forecast Horizon defines the temporal granularity and total look-ahead. Active Security Signals compresses the number of entities/events requiring attention.

4. Risk score explanation

The 0–100 score is presented as a normalized operational risk index. In a production system it should be produced from calibrated model outputs or a documented ensemble of anomaly, behavioral and threat signals. The demo value of 86% is synthetic. The +23% indicator represents change versus a notional baseline.

5. Forecast explanation

The risk chart uses a solid cyan observed line and an orange dashed forecast line. Hovering exposes time, risk and confidence. Clicking a point opens the exact forecast window and predicted stage. Confidence should be interpreted as model certainty, not probability of compromise unless the backend explicitly calibrates it that way.

6. Attack progression

The demonstration predicts Credential Access, then Lateral Movement, then Command & Control. Each stage has a confidence and time window. Clicking a stage opens investigation guidance. The UI intentionally labels the sequence as predicted rather than confirmed.

7. Network graph

React Flow provides pan, zoom, fit and reset controls. Nodes represent workstations, servers, databases, file servers and suspected hosts. Edges represent network flows. Cyan/blue indicates normal traffic, red suspicious traffic and orange dashed paths predicted traffic. Node clicks reveal IP, hostname, role, risk, traffic volume, connections and state. Edge clicks reveal source, destination, protocol, port, traffic and risk.

8. SHAP explainability

The feature panel ranks positive contributions: SYN Activity +0.28, Unique Hosts +0.21, Port Diversity +0.18, IAT Mean Deviation +0.12, Flow Count +0.09, Retransmissions +0.06, Bytes per Flow +0.04 and Packet Size Variance +0.03. Clicking a feature exposes its description and risk implication. These values are illustrative SHAP-style contributions.

9. MITRE ATT&CK;

The top mapping is T1021 Remote Services at 72%, followed by T1047 Windows Management Instrumentation at 65% and T1078 Valid Accounts at 58%. A technique drawer shows ID, name, tactic, confidence, why it was mapped and supporting evidence. Production mappings should preserve provenance and model/version metadata.

10. Alerts

Alerts contain ID, timestamp, severity, source IP, destination IP, description, stage, confidence and status. The Alerts screen supports search plus severity/status filtering. Clicking an alert opens an investigation drawer. Demo statuses are Open, Investigating, Monitoring and Resolved.

11. Upload workflow

The Ingest telemetry control accepts PCAP, PCAPNG and CSV. The demo validates the extension, waits briefly, then reports a simulated staging result. In production, the browser would upload the file to a secure ingestion endpoint, where parsing, normalization and feature extraction occur server-side.

12. Forecasting workflow

Stage	Responsibility
Telemetry	PCAP/CSV/network sensors produce raw events.
Preprocessing	Validate, normalize and window traffic.
Feature extraction	Calculate flow, timing, host and protocol features.
Risk model	Estimate current anomaly/attack risk.
Forecast engine	Predict future risk windows.
Stage classifier	Estimate the next attack stage.
Explainability	Generate feature contributions.
MITRE mapping	Map predicted behaviors to ATT&CK; techniques.
Dashboard	Present evidence for analyst action.

13. Example attack scenario

A suspected workstation 192.168.1.5 begins generating elevated SYN activity and contacts multiple internal hosts. Port diversity and unique-host count increase. The forecast raises current risk to 86% and predicts lateral movement at 82%

confidence. The graph highlights an RDP path toward 192.168.1.10 and a predicted path toward 192.168.1.25. T1021 Remote Services becomes the leading ATT&CK; hypothesis. The analyst opens the linked alert and validates the evidence.

14. What each visualization means

KPI cards summarize state; the trajectory shows time; the stage timeline shows sequence; the graph shows relationships; SHAP shows feature influence; MITRE shows standardized behavioral interpretation; alerts show operational workflow. The intended reading order is from broad risk to specific evidence.

15. How to interpret the dashboard

Start with current risk, then ask whether the forecast is rising. Identify the predicted stage. Inspect the strongest contributors. Find the highest-risk host and predicted path. Check ATT&CK; mapping and supporting evidence. Finally open the alert and decide what must be validated. Do not treat a forecast as proof of compromise.

16. Limitations of the frontend demo

There is no live packet capture, trained forecasting model, production calibration, authentication system, persistent database, or real alert transport. Upload processing is simulated. Network and SHAP values are deterministic mock values. These limitations are intentional for a presentation frontend and are clearly represented in the UI.

17. Future backend integration

The primary integration boundary is `src/services/forecastService.ts`. Replace the mock service with typed HTTP calls while retaining the `ForecastSnapshot` interface. Recommended endpoints include `GET /v1/forecast`, `POST /v1/telemetry/ingest`, `GET /v1/network`, `GET /v1/alerts`, `GET /v1/mitre/mappings` and `GET /v1/evidence/{eventId}`. Add authentication, authorization, input validation, audit logging, rate limits, secure file handling and TLS.

Evaluator takeaway

The key product concept is not another SOC screen that reports what happened. It is a decision-support layer that links current telemetry to future risk, predicted attack progression, explainability, attack-path context and standardized ATT&CK; interpretation. The frontend is structured so a real forecasting backend can replace mock data without rebuilding the interface.